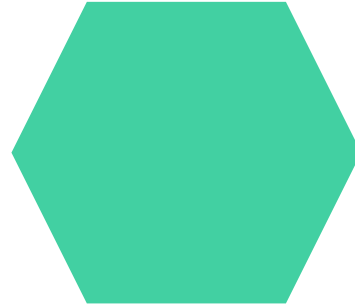
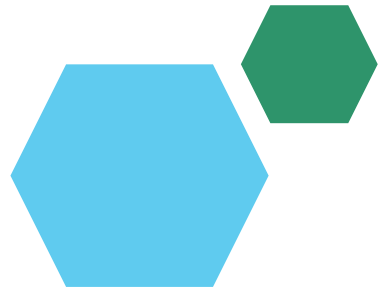


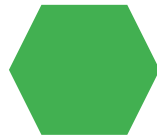


Project Title - **Seasonal Agriculture Performance Analysis**

Student Name - Udipti Nagar

College Name - Sagar Institute of Research and Technology

AICTE STU ID : STU684af1354fd1a1749741877



PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.

However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.

The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description



Seasonal Agriculture Performance Analysis is a data analytics project that examines agricultural activities across different seasons, crops, geographical areas, and farming conditions.

The project analyzes environmental factors, farming practices, resource usage, crop yield, production, and economic performance to identify meaningful seasonal patterns, trends, relationships, and variations.

The analysis uses data cleaning, statistical analysis, visualization, correlation analysis, and seasonal comparisons to understand differences in agricultural performance. The findings are used to develop data-driven insights and recommendations for better seasonal agricultural planning.



WHO ARE THE END USERS?



End Users: Farmers, Agricultural Planners & Government Departments, Agricultural Researchers & Data Analysts, Agricultural Consultants/ Agronomists, and Farm Management Organizations.

The project provides these stakeholders with data-driven insights into seasonal crop performance, resource usage, environmental conditions, profitability, and agricultural trends to support better planning and decision-making.



Technology Used

- **Python** – Main programming language for data analysis
- **Google Colab** – Cloud-based environment for developing and executing the project
- **Pandas** – Data loading, cleaning, manipulation, and aggregation
- **NumPy** – Numerical computations and data processing
- **Matplotlib** – Data visualization and plotting
- **Seaborn** – Statistical and comparative visualizations
- **SciPy** – Statistical analysis and hypothesis testing
- **CSV** – Dataset/file format
- **GitHub** – Project code and notebook version control/storage



RESULTS

 AgricultureDataAnalysis.ipynb ☆ ☁

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```
[ ] print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

✓
Number of rows: 4000
Number of columns: 28

✓**4. Top 5 rows**

```
[ ] df.head()
```

✓

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	24.3	69.1	5.8	...	...
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	25.7	78.4	5.1	...	...
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6	23.9	56.4	10.3	...	...
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.2	29.7	65.2	6.3	...	...
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6	34.1	55.2	6.5	...	...

5 rows x 28 columns

6

RESULTS

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▼ 6. Dataset Shape

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print("Dataset Shape:", df.shape)

▼ Dataset Shape: (4000, 28)

▼ 7. Dataset Information

[]

▶ df.info()

▼
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
Column Non-Null Count Dtype

0 Farm_ID 4000 non-null object
1 State 4000 non-null object
2 District 4000 non-null object
3 Crop 4000 non-null object
4 Season 4000 non-null object
5 Farm_Area_Hectares 4000 non-null float64
6 Rainfall_mm 3952 non-null float64
7 Avg_Temperature_C 4000 non-null float64
8 Humidity_pct 4000 non-null float64
9 Sunlight_Hours_Day 4000 non-null float64
10 Soil_pH 4000 non-null float64
11 Soil_Moisture_pct 3960 non-null float64
12 Nitrogen_kg_ha 4000 non-null float64

RESULTS

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▼ 10. Missing value analysis

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```
missing_values = df.isnull().sum()

missing_values = missing_values[missing_values > 0].sort_values(ascending=False)

print("Columns containing missing values:")
display(missing_values)
```

▼

Columns containing missing values:

	0
Rainfall_mm	48
Soil_Moisture_pct	40
Yield_Tonnes_Ha	32

dtype: int64

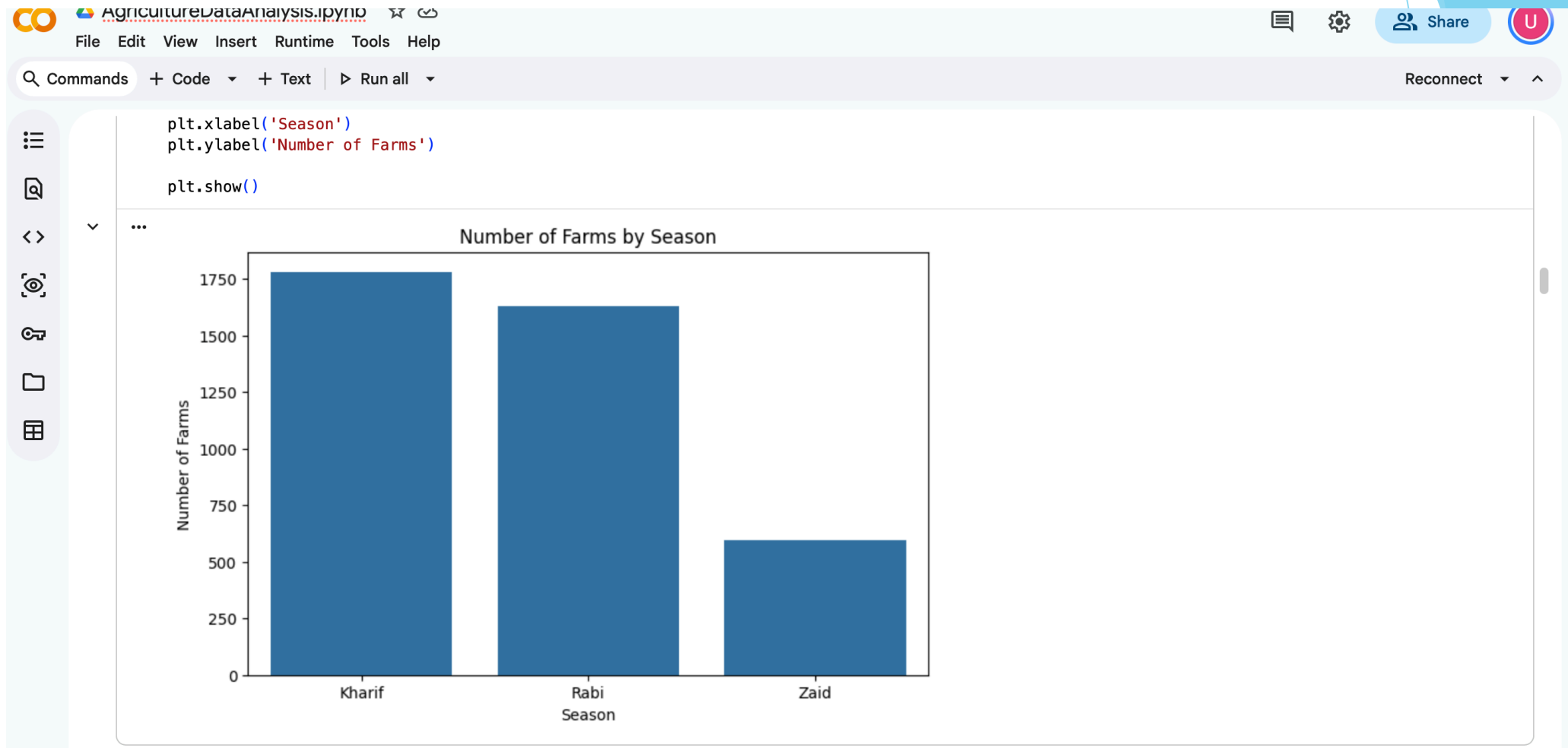
▼ 11. Missing-value percentage

[]

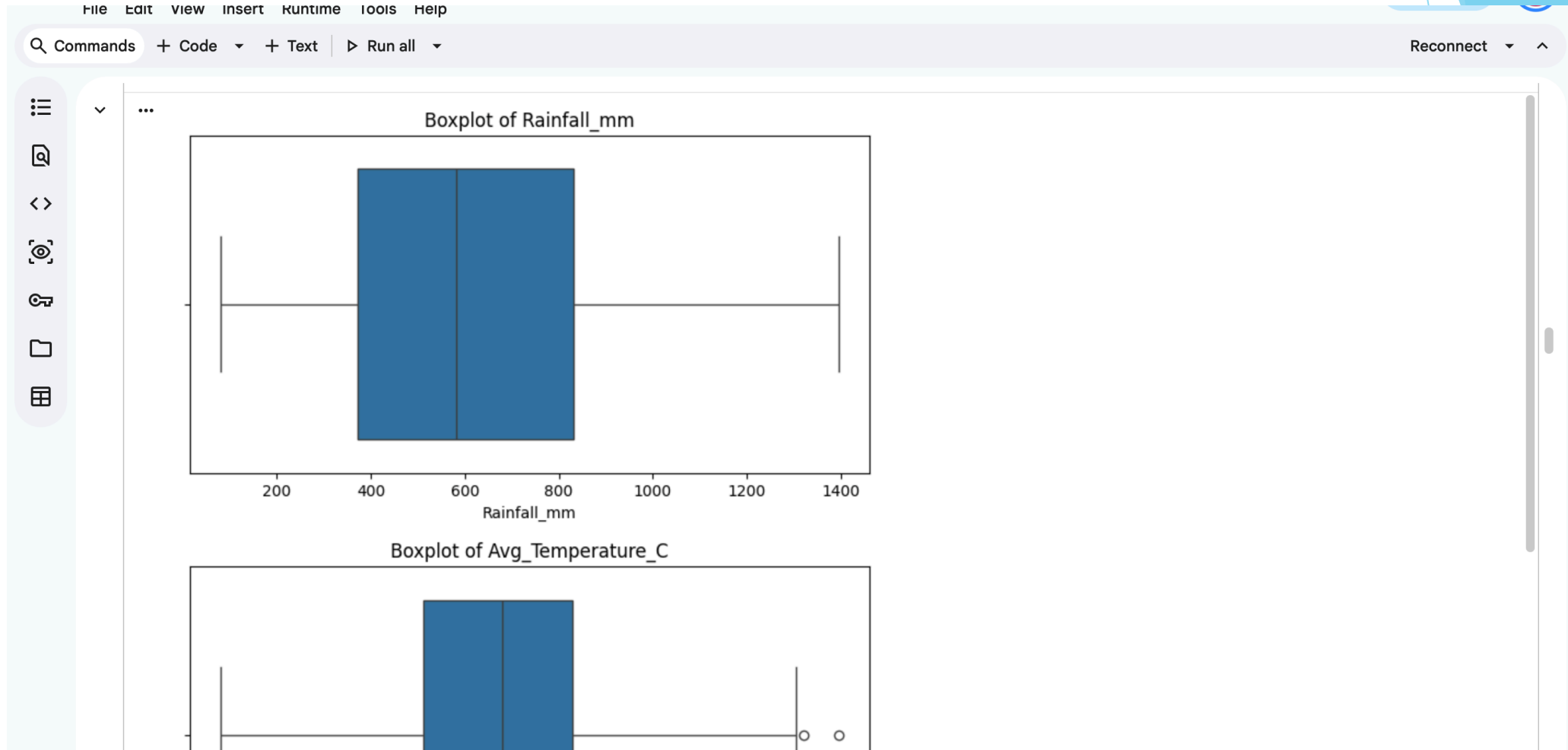
```
missing_percentage = (df.isnull().sum() / len(df)) * 100

missing_table = pd.DataFrame({
    'Missing Values': df.isnull().sum(),
    'Missing Percentage': missing_percentage
})
```

RESULTS



RESULTS



Future scope



The project can be extended by integrating real-time weather data, machine learning-based yield prediction, crop recommendation systems, resource optimization, disease and pest prediction, larger geographical datasets, and interactive dashboards.

These enhancements can transform the current analytical study into a more comprehensive agricultural decision-support system.



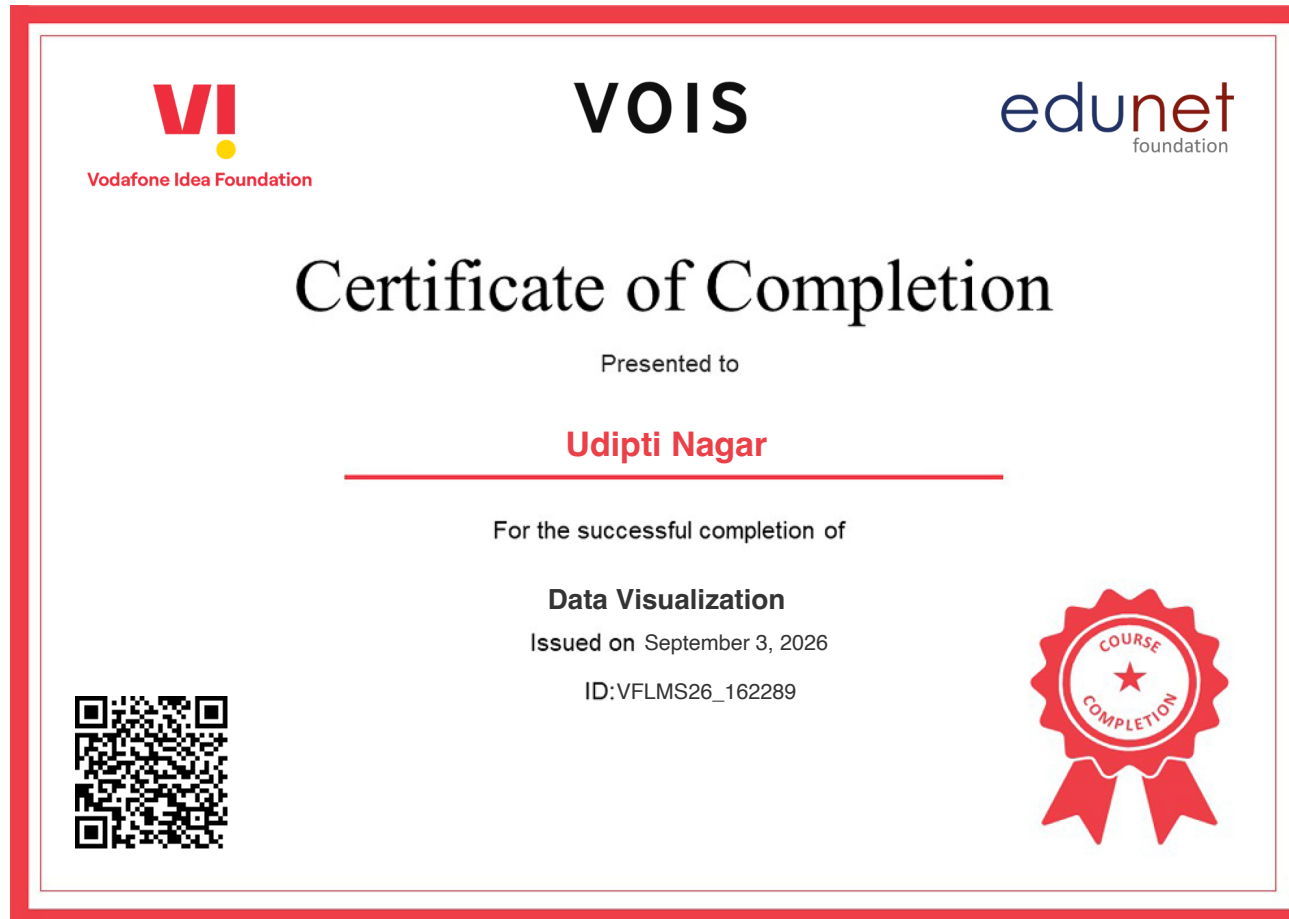
GitHub Link



<https://github.com/udiptinagar/Major-Project---Agriculture-Data-Analysis>



VOIS Course completion certificate - Course name- Data Visualization



Thank you

